

Work Breakdown for Device Emulator Intern:

Overview:

The relevant pieces have nearly been completed to MVP, and should not change once that is the case. Speed is an utmost priority as the device emulator will serve as a method of testing the full system, and as such time is needed to design and run tests or fix as need be.

1. Dates & Schedule:

- a. This position is expected to begin on **June 1, 2026** and go until **August 5, 2026** (end of the semester)
- b. Work is expected to be completed during all 5 remaining sprints. Here is a vague breakdown, timeline wise:
 - i. Sprint 1: Onboarding, learning spike of the tech stack(specifically AWS), initial documentation and artifacts(uml, sequence diagrams, any other architecture) for the base CLI tool created and approved by team
 - ii. Sprint 2-3: CLI emulator development started and a prototype of the software is completed. Should be able to connect dummy devices to the system and send data as those devices
 - iii. Sprint 4: Final touchups to the CLI version, and start of converting the tool to a GUI variant
 - iv. Sprint 5: GUI tool brought to a working and bug free state

2. Processes + tasking:

- a. The tool should be kept in its own github repository, and proper branching and versioning procedures to maintain a clean and comprehensible workflow should be followed.
- b. The main branch will represent the current main product, while additional branches should be work in progress features.
 - i. Any merges into the main branch should be done through a pull request, which requires at least one other team member to view and approve before merging.

3. Device Emulator Tool. This includes:

- a. A full set of all **software engineering artifacts** needed to plan out the architecture of the tool. Each must be approved by the team and modified accordingly based on feedback before implementation is allowed to begin. These include:
 - i. Full UML layout of the application
 - ii. Sequence diagrams for program flow
 - iii. Data model diagrams for data storage
 - iv. A breakdown of the tool into smaller story sized tasks, with story point sizes attached
 - v. A clear schedule of what tasks will be completed during which sprint
- b. A **command line tool** that emulates the full device functionality on a local computer, and communicates with the real server. The tool must run both locally

and must be capable of being run via github actions when integrated into the testing workflow. CLI functionality includes:

- i. New device registration
 - ii. Trail data uploads
 - iii. Device certificate renewal
 - iv. Device expiration and new certificate creation
- c. A **graphical user interface** version of the CLI tool. This is simply porting the tool to a GUI variant, so that it can be easily used by a non technical individual to demonstrate and verify functionality of the full system
- d. **Full documentation** both in separate documents along with clean code, comments, and docstrings. The emulator will additionally be used to provide clear guidance to the device firmware team, who will be implementing the same functionality onto the actual devices this fall. We will not be around to assist, so everything must be explicitly clear so they can examine the emulator, understand it, and replicate functionality in their own firmware code without assistance.
4. Supporting the team. This is a very ambitious senior project - there is no doubt that support will be needed in other areas. Some tasks that can be expected are:
- a. Aiding the main team, most likely with integrating the CLI tool into the testing workflow and writing tests for the main system
 - b. Providing feedback on completed work from the team
 - c. Participating in design decisions
 - d. Attending meetings and actively being involved and a contributing member of the Trailblazers